Vaishak D Karkera

Embedded Systems — IoT — Distributed Sensing Systems

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Education

Bachelor of Engineering in Electronics and Communication Engineering 2023 – 2027
Sahyadri College of Engineering and Management, VTU CGPA: 8.8 / 10.0

Technical Skills

Programming: C, C++, Python, Embedded C, Verilog
Embedded Platforms: ESP32, ESP8266, Arduino, STM32
Communication Protocols: UART, SPI, I2C, MQTT, HTTP, WebSockets
Firmware & Interfacing: GPIO, Timers, Interrupts, Sensor Interfacing, I2S Communication
IoT & Cloud: Firebase Realtime Database, Real-time Data Logging, Cloud-based Monitoring
Simulation & Visualization: Firebase Dashboards, Real-time Monitoring Interfaces, Sensor Data Visualization
ML & Vision: OpenCV, MediaPipe, YOLOv5
Hardware & Design: PCB Design, Cadence Virtuoso, Scilab
Tools: Arduino IDE, ESP-IDF, PlatformIO, Git, GitHub, VS Code, Linux CLI
Core Knowledge: Real-Time Systems, IoT Architecture, Distributed Monitoring Systems, Digital Electronics, Operating Systems

Selected Engineering Projects

Distributed Swarm-Based Environmental Surveillance System 2025 – Present

- Developing distributed ESP32-powered autonomous sensing system using multiple cooperative nodes for continuous environmental monitoring and intelligent threat detection.
- Designed multi-agent coordination architecture enabling real-time communication, adaptive patrol routing, and scalable distributed sensing across dynamically changing environments.
- Integrated ultrasonic sensing, GPS tracking, environmental monitoring, wireless alert generation, and cloud-connected monitoring interfaces for fault-tolerant surveillance operations.
- Implemented distributed multi-node monitoring architecture focused on scalable autonomous sensing, cooperative detection, and adaptive environmental awareness.

Smart Public Distribution System 2025

- Engineered automated ration dispensing system using ESP32 and HX711 load cell achieving ±5g measurement accuracy for transparent quantity allocation.
- Implemented sensor-driven dispensing, real-time weight verification, and Firebase-based cloud transaction logging to reduce manual leakage and improve auditability.
- Designed real-time monitoring dashboard with cloud-synchronized transaction visibility and live operational updates.
- Built secure role-based web interface for Admin, Distributor, and Beneficiary with duplicate prevention using unique ID validation mechanisms.
- Selected in **Smart India Hackathon 2025 Internal Evaluation Round**.

AI-Enabled Tamper Detection System for Weighing Machines 2025

- Developed embedded tamper detection system for electronic weighing machines using sensor monitoring and anomaly detection techniques.
- Implemented real-time detection pipeline to identify unauthorized manipulation, abnormal measurement variations, and operational inconsistencies.
- Designed firmware and intelligent alert mechanisms for secure, transparent, and reliable weighing operations.

Smart Inventory & Shelf Management System — Top 10 Finalist 2025

- Developed IoT inventory tracking system integrating ESP32, RC522 RFID, and HX711 load cell for multi-layer product verification.
- Implemented RFID-based product identification over SPI with weight validation achieving 90%+ stock confirmation accuracy.
- Integrated YOLOv5 object detection for visual verification and delivered fully functional prototype within 24-hour hackathon constraint.

Achievements & Certifications

- Smart India Hackathon (SIH) 2025 – Selected in Internal Evaluation Round
- Winner – Reality Rewritten Event, Udbhava 2025

- Top 10 Finalist – Smart Inventory & Shelf Management, Skill Sangam 2K25 IoT Hackathon
- Samsung RISC-V Talent Development Program (6 Weeks) – RISC-V ISA, Assembly Programming, VSD Squadron Mini
- Infosys Springboard – Internet of Things Certification
- Infosys Springboard – Python Programming Certification
- Workshop on Radar and LiDAR Systems for ADAS Applications
- Participant – HacXerve 2.0 Hardware Hackathon
- Participant – IEEE CEDA IDEATHON 2025